Fresnel reflection modeling within the higher-order parabolic equation and discrete nonlocal boundary conditions

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***Abstract*—This paper is devoted to modeling the interface between two media with different dielectric properties. Special attention is paid to the Earth's surface modeling in the tropospheric radio-wave propagation problem. The finite-difference approximations of the one-way Helmholtz equation are used. Instead of the local Leontovich boundary condition, a discrete nonlocal boundary condition is used. In contrast to the Leontovich boundary condition, the proposed approach does not impose restrictions on the propagation angle and the dielectric permittivity of the underlying surface. The proposed numerical scheme allows using an arbitrary Padé approximation order for the longitudinal propagation operator and a fourth-order approximation for the transverse operator. The results of the numerical modeling demonstrate the effectiveness of the proposed approach.**

***Index Terms*—nonlocal boundary condition, Fresnel reflection coefficient, Padé approximation, Parabolic equation, Helmholtz equation**

I. Introduction

TROPOSPHERIC radio-wave propagation is greatly affected by the dielectric properties of the ground. It is common to use the Leontovich impedance boundary condition (BC) [1], [2] to take into account the ground properties within the parabolic equation (PE) method. Leontovich BC is an approximate one and may be applied under the following two conditions

$$\varepsilon_g \gg 1, \tag{1}$$

$$\theta \approx 0, \tag{2}$$

where ε_g is the complex-valued permittivity of the ground and θ is the grazing angle. Condition (2) is met for the long range radio-wave propagation, when the propagation involves only the waves directed almost horizontally. In other cases, the significant dependence of the reflection coefficient on the incidence angle should be considered. The fulfillment of condition (1) depends on the dielectric properties of the underlying surface. For example, it performs well for the sea surface ($\varepsilon_g = 69.1+42.9i$ when frequency is equal to 3 GHz), but requires refinement for the dry ground ($\varepsilon_g = 3 + 0.01i$ for the same frequency). Widely used split-step Fourier (SSF) method with mixed Fourier transform may yield a divergence of the resulting numerical solution in some important propagation scenarios [3], [4]. The point is that in addition to the above-mentioned restrictions, this method is incompatible with transparent BC and zero reflection coefficient. The zero-valued reflection coefficient occurs when vertically polarized waves incident at the Brewster angle [5]. Another example of a near-zero reflection coefficient is the Miller-Brown model for a rough sea surface [6]. Modifications of the SSF method to account for the angle-dependent BC were considered in [3], [5], but they have not yet found widespread use. Thus, under certain conditions, the existing approaches to model the properties of the underlying surface in the PE method require clarification.

There are more and more studies devoted to the application of the finite-difference higher-order PE [7] to the tropospheric radio-wave propagation problems [8], [9], [10]. This is primarily due to the need to correctly account for the complex BC. The finite-difference nature of this method allows for more efficient handling of complex BC. In particular, this method can be used to effectively model transparent BC designed to artificially reduce the computational domain when solving problems in unbounded domains [11], [12], [13]. Transparent boundaries in the PE method are modeled using nonlocal BC [14], [15], [16], [17]. Previously, it was shown how to use the split-step Padé method with discrete transparent BC for over-horizon radio-wave propagation in a tropospheric waveguide [10], [18]. In the computational hydroacoustics, the method of nonlocal BC was used to account for the rough sea surface and the water-ground boundary [19], [20].

In this paper, the nonlocal BC for the split-step Padé method is generalized to the case of semi-transparent BC, which is essentially the interface between two media.

II. Problem statement

The wave propagation process in the considered problem is described by a two-dimensional scalar Helmholtz equation [2]

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial z^2} + k^2 m^2(x,z)\psi = 0, \tag{3}$$

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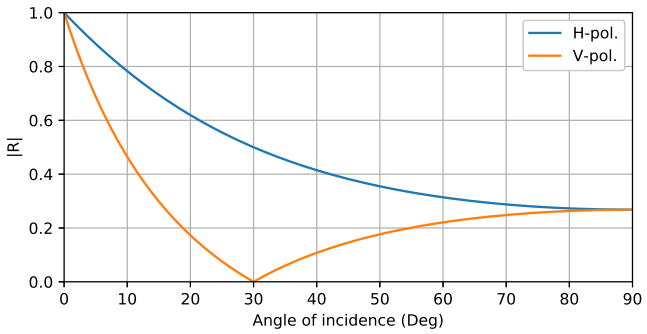


Fig. 1. Dependence of the reflection coefficient on the grazing angle θ when $\varepsilon_g = 3$.

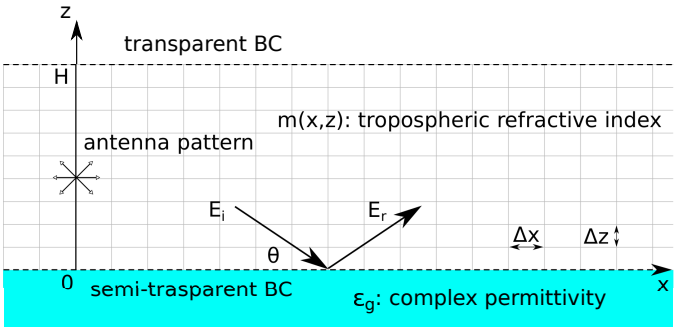


Fig. 2. Schematic description of the considered problem.

where $\psi(x,z)$ is the complex-valued wavefield, $m^2(x,z)$ is the modified tropospheric refractive index, k is the wavenumber. The solution is sought on the domain $\Omega = [0,+\infty) \times [0,H]$. A transparent BC with linearly increasing modified refractive index is set at $z = H$ [21]. It is further assumed, without limiting generality, that $m(x,z) \equiv 1$ in some small neighborhood of $z = 0$. Domain $\Omega_g = (-\infty,+\infty) \times (-\infty,0]$ is uniformly filled with a homogeneous medium with complex permittivity ε_g. The main task of this study is to determine the lower BC at $z = 0$, which can take into account an arbitrary complex permittivity of the ground ε_g.

Fresnel reflection coefficients for a surface with complex permittivity ε_g for horizontal and vertical polarization are expressed through horizontal $k_x = \cos\theta$ and vertical $k_z = \sin\theta$ wavenumbers as follows [5]:

$$R_h(k_x) = \Gamma_h(k_z) = \frac{k_z - \sqrt{k^2\varepsilon_g - k_x^2}}{k_z + \sqrt{k^2\varepsilon_g - k_x^2}},$$

$$R_v(k_z) = \Gamma_v(k_z) = \frac{\sqrt{k^2\varepsilon_g - k_x^2} - \varepsilon_g k_z}{\sqrt{k^2\varepsilon_g - k_x^2} + \varepsilon_g k_z}.$$

The dependence of the reflection coefficient on the grazing angle θ when $\varepsilon_g = 3$ is depicted in Fig. 1. In this paper, we will be particularly interested in the case of vertical polarization and Brewster angle incidence θ_B, such that $\Gamma_v(k_z = k\sin\theta_B) = 0$. As previously mentioned, this case is the most difficult for the existing PE algorithms. In the example shown $\theta_B = 30°$. The Brewster angle is generally calculated using the formula $\theta_B = \arcsin\left(1/\sqrt{\epsilon_g + 1}\right)$.

The wave field is generated by the initial condition at $x = 0$. The schematic description of the considered problem is depicted in Fig. 2.

III. Approximation of the Propagation Operator

Denote $u_j^n = u(\Delta x, \Delta z)$ and $u^n(z) = u(\Delta x, z)$. Neglecting to backscatter, the step-by-step solution of (3) can be written using the pseudo-differential propagation operator [22] as follows [2]

$$u^{n+1} = \exp\left(ik\Delta x\left(\sqrt{1+L}-1\right)\right)u^n, \tag{4}$$

$$u(x,z) = e^{-ikx}\psi(x,z),$$

$$Lu = \frac{1}{k^2}\frac{\partial^2 u}{\partial z^2} + \left(m^2 - 1\right)u.$$

Using the definition of the pseudo-differential operator [22], expression (4) can be overwritten using Fourier transformed function $\tilde{u}^n$ as follows

$$\tilde{u}^{n+1} = \exp\left(ik\Delta x\left(\sqrt{1+s}-1\right)\right)\tilde{u}^n, \tag{5}$$

$$s = -\left(\frac{k_z}{k}\right)^2 + \left(m^2(x,z) - 1\right),$$

$$\tilde{u}^n(k_z) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} u^n(z)e^{-ik_z z}dz.$$

Next, we apply the Padé approximation [7], [23] to the propagation operator in (5)

$$\exp\left(ik\Delta x\left(\sqrt{1+s}-1\right)\right) \approx \prod_{l=1}^{p}\frac{1+a_l s}{1+b_l s}. \tag{6}$$

Substitute approximation (6) into (5). By introducing p-1 auxiliary functions $v_1^n(z), v_2^n(z), \ldots, v_{p-1}^n(z)$ and applying the inverse Fourier transform, the action of the propagation operator (4) may be rewritten as a system of similar one-dimensional differential equations that are solved sequentially using the Crank-Nicholson scheme

$$\begin{cases} (1+b_1 L)\,v^n = (1+a_1 L)\,u^{n-1} \\ (1+b_l L)\,v_l^n = (1+a_l L)\,v_{l-1}^n \qquad l = 2,\ldots,p-1. \\ (1+b_1 L)\,u^n = (1+a_1 L)\,v_{p-1}^n \end{cases} \tag{7}$$

The second-order differential operator is approximated by the fourth-order scheme [24]

$$\frac{\partial^2}{\partial z^2} \approx \frac{1}{\Delta z^2}\frac{\delta^2}{1+1/12\delta^2}, \tag{8}$$

where δ^2 is a second difference operator, which is defined as follows

$$\delta^2 u_j = u_{j+1} - 2u_j + u_{j-1}.$$

IV. Nonlocal boundary condition

Applying $\mathcal{Z}$-transformation

$$\mathcal{Z}\{u^n\} = \hat{u}(\xi) = \sum_{n=0}^{\infty} \xi^{-n} u^n$$

to (6) along the longitudinal coordinate n leads to a semi-discrete dispersion relation

$$\xi = \prod_{l=1}^{p} \frac{1 + a_l s}{1 + b_l s}. \tag{9}$$

Consider (9) as a polynomial of variable s. Taking into account that $s = -k_z^2/k^2$ when $z = 0$, it is well seen that each value of ξ corresponds to p different values of the vertical wavenumber k_z.

Now let's go back to the spatial area by z. Following [21], $\mathcal{Z}$-transformed scheme (7) in the vicinity of $z = 0$ can be rewritten in the following vector form

$$\frac{\partial^2}{\partial z^2}\hat{\boldsymbol{\chi}}(z) = k^2 \mathbf{J}\hat{\boldsymbol{\chi}}(z), \tag{10}$$

where

$$\hat{\boldsymbol{u}}(z) = \mathbf{P}\hat{\boldsymbol{\chi}}(z),$$

$$\hat{\boldsymbol{u}}(z) = \begin{pmatrix} \hat{v}_1(z) & \dots & \hat{v}_{p-1}(z) & \hat{u}(z) \end{pmatrix}^T,$$

matrices $\mathbf{P}$ and $\mathbf{J}$ are the solutions of the generalized eigenvalue problem

$$\mathbf{B}(\xi) = \mathbf{A}(\xi)\mathbf{P}(\xi)\mathbf{J}(\xi)\mathbf{P}^{-1}(\xi),$$

where

$$\mathbf{A}(\xi) = \begin{pmatrix} \xi b_1 & & & & -a_1 \\ -a_2 & b_2 & & & \\ & \ddots & \ddots & & \\ & & -a_{p-1} & b_{p-1} & \\ & & & -a_p & b_p \end{pmatrix},$$

$$\mathbf{B}(\xi) = \begin{pmatrix} -\xi & & & & 1 \\ 1 & -1 & & & \\ & \ddots & \ddots & & \\ & & 1 & -1 & \\ & & & 1 & -1 \end{pmatrix}.$$

Diagonal matrix $\mathbf{J} = diag\{s_1, s_2 \dots s_p\}$, consists of the roots of characteristic polynomial (9).

Each row of the system (10) consists of p independent equations of the form

$$\frac{\partial^2}{\partial z^2}\chi_l = k^2 s_l \chi_l. \tag{11}$$

General solution of (11) may be written as follows

$$\chi_l(z) = c_1 \exp\left(-ik\sqrt{-s_l}z\right) + c_2 \exp\left(ik\sqrt{-s_l}z\right).$$

The first and second terms correspond to the incident and reflected plane waves respectively. Given the presence of a reflecting surface with the reflection coefficient $\Gamma(k_z)$, the coefficients c_1 and c_2 are related by the ratio

$$c_2 = \Gamma(k_{z,l}(\xi))\, c_1.$$

Vertical wavenumber $k_{z,l}(\xi)$ is expressed through s_l as follows

$$k_{z,l}(\xi) = k\sqrt{-s_l(\xi)}.$$

Using approximation (8), one can write down the discrete variant of (11)

$$\delta^2 \chi_{l,j} = \Delta z^2 k^2 s \left(1 + 1/12\delta^2\right) \chi_{l,j}. \tag{12}$$

General solution of (12) takes the following form

$$\chi_{l,j} = c_1 \mu_l^{-j} + c_2 \mu_l^j,$$

where $|\mu| < 1$ is obtained by solving the corresponding quadratic equation. To obtain a discrete nonlocal BC at $z = 0$, the following relation should be considered

$$\frac{\chi_{l,0}}{\chi_{l,1}} = \frac{1 + \Gamma(k_{z,l})}{\mu_l^{-1} + \Gamma(k_{z,l})\,\mu_l}.$$

Then the discrete nonlocal BC can be written as follows [10]

$$\boldsymbol{u}_{J-1} - \mathbf{D}^0 \boldsymbol{u}_J^n = \sum_{m=1}^{n-1} \mathbf{D}^m \boldsymbol{u}_J^{n-m},$$

where coefficients of BC are determined from the inverse $\mathcal{Z}$-transform

$$\mathbf{D}^m = \frac{\tau^m}{2\pi} \int_0^{2\pi} \hat{\mathbf{D}}\left(\tau e^{i\varphi}\right) e^{im\varphi} d\varphi,\ \tau > 1, \tag{13}$$

$$\hat{\mathbf{D}}(\xi) = \mathbf{P} diag\left\{\frac{\chi_{1,0}}{\chi_{1,1}}, \frac{\chi_{2,0}}{\chi_{2,1}}, \dots, \frac{\chi_{p,0}}{\chi_{p,1}}\right\} \mathbf{P}^{-1}.$$

An adaptive version of the Filon-Clenshaw-Curtis quadrature [25] is used for the numerical calculation of the integral (13).

It is important to note that the k_z used are essentially complex-valued. Thus, the reflection coefficient Γ must has an analytical continuation to the complex plane.

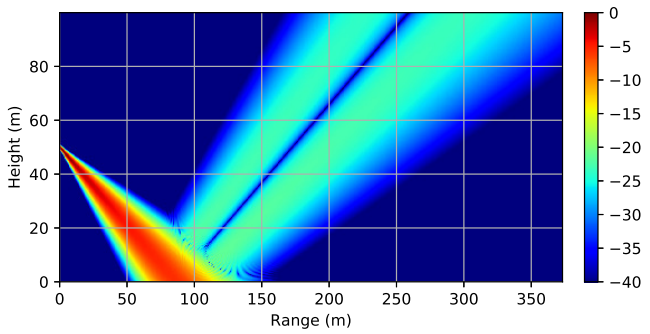


Fig. 3. Amplitude of the field $(10 \lg |\psi(x,z)|)$, obtained by the proposed approach.

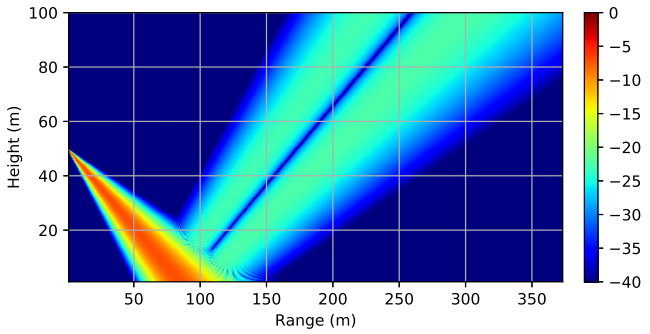


Fig. 4. Amplitude of the field $(10 \lg |\psi(x,z)|)$, obtained by the two-ray model.

V. Numerical Results and Discussion

Comparative analysis of the proposed approach is carried out with a simple two-ray model [26] with angle-dependent lower boundary and the SSF method, implemented in the PEOOL software [27]. We consider the incidence of a vertically polarized Gauss beam [2] to a surface with a complex permittivity $\varepsilon_g = 3$. This surface roughly corresponds to dry ground. The radiation frequency is 3 GHz, beam width is 5° and the elevation angle is -30°, which is equal to the Brewster angle for a given surface. The source is located at the height of 50 m above the surface. The dependence of the reflection coefficient of the specified surface on the grazing angle is shown in Fig. 1. Fig. 3, 4 and 5 represent the results of numerical simulation obtained by the Padé method with nonlocal BC, two-ray model, and SSF method, respectively. The results obtained by the first two methods actually coincide, while the figures show a characteristic zero field, which corresponds to the Brewster angle incidence. As expected, the SSF method with Leontovich BC failed to correctly model the reflection. The field amplitude at the height of 50 m obtained by all mentioned methods depicted in Fig. 6. It is clear that the proposed method correctly takes into account the dependence of the reflection coefficient on the incidence angle. The CPU time of the considered methods does not differ significantly, and the need for preliminary calculation of nonlocal BC coefficients (13) does not significantly affect the speed of calculations.

VI. Conclusion

The nonlocal BC method was applied to account for the Earth's surface parameters. Any Fresnel reflection coefficient may be modeled within the proposed approach. In general, the proposed method allows one to take into account an arbitrary refractive index of the troposphere. Although the latter usually appears for small propagation angles, i.e. when the local Leontovich BC is applicable. It should be noted that the described mismatches with the SSF method are characteristic of vertical polarization. All three considered methods yield very similar results for horizontal polarization.

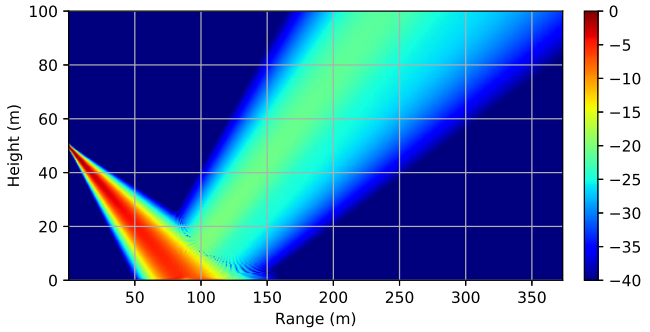


Fig. 5. Amplitude of the field $(10 \lg |\psi(x,z)|)$, obtained by the SSF method with Leontovich BC.

The proposed method can be used in other subject areas where the PE method is used, for example, in hydroacoustics and optics. The method can be easily applied to modeling the propagation of millimeter waves in the framework of 5G technology.

It is theoretically possible to model any reflection coefficient within the proposed approach. In particular, this applies to the Miller-Brown model. However, this requires a more detailed study of the analytical continuation of an arbitrary refractive index to the complex plane. This problem can be solved in future works. Another promising direction is the generalization

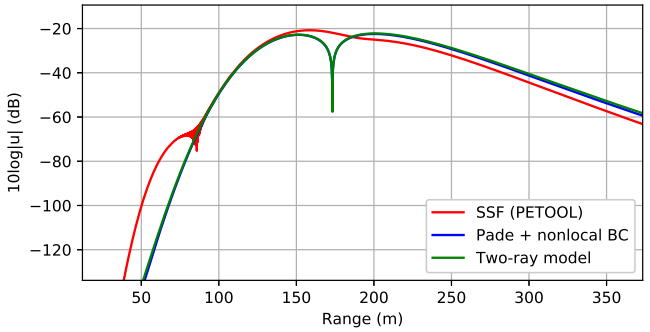


Fig. 6. Amplitude of the field $(10 \lg |\psi(x,z)|)$ at the height $z = 50$.

of the obtained nonlocal BC to a three-dimensional parabolic equation.

The proposed method is implemented in an open-source Python 3 library [28].

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